

E.3 / PAPER 1 / QUESTION 27

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27. The table shows how the count rate from a sample of a radioactive nuclide varies with time t . The nuclide has a half-life of 50s. The average background count rate is constant.

t/s	Count rate/ s^{-1}
0	72
50	40

What count rate will the detector measure when $t = 150\text{ s}$?

- A. 8 s^{-1}
- B. 16 s^{-1}
- C. 24 s^{-1}
- D. 32 s^{-1}